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Course:DSA0602-Data handling and visuilization

**EXPERIMENT 1: Monthly Sales Analysis**

**Step 1: Load Package**

```
library(ggplot2)
library(plotly)
```

Step 2: Create Dataset

```
sales <- data.frame(
  Month =
  c("Jan","Feb","Mar","Apr","May"),
  Sales = c(1200,1500,1800,1600,2000)
)

print(sales)
```

Step 3: Line Chart

```
ggplot(sales, aes(x = Month, y = Sales, group =
  1)) + geom_line(color = "blue", linewidth =
  1.2) + geom_point(color = "red", size = 3) +
  labs(
    title = "Monthly Sales
    Trend", x = "Month",
    y = "Sales"
```

```
 ) +
 theme_minimal()
```

Step 4: Bar Chart

```
ggplot(sales, aes(x = Month, y = Sales, fill = Month)) +
 geom_bar(stat = "identity") +
 labs(
  title = "Monthly Sales
  Comparison", x = "Month",
  y = "Sales"
 ) +
 theme_minimal()
```

Step 5: Scatter Plot

```
ggplot(sales, aes(x = Month, y = Sales))
 + geom_point(size = 4, color = "blue")
 +
 geom_smooth(aes(group = 1), method = "lm", se = FALSE, color =
 "red") + labs(
  title = "Monthly Sales Scatter
  Plot", x = "Month",
  y = "Sales"
 ) +
 theme_minimal()
```

Step 6: Interactive Dashboard

```
p1 <- plot_ly(sales,
       x =
       ~Month, y
       = ~Sales,
       type = "scatter",
       mode = "lines+markers")
```

p2 <- plot_ly(sales,

x =

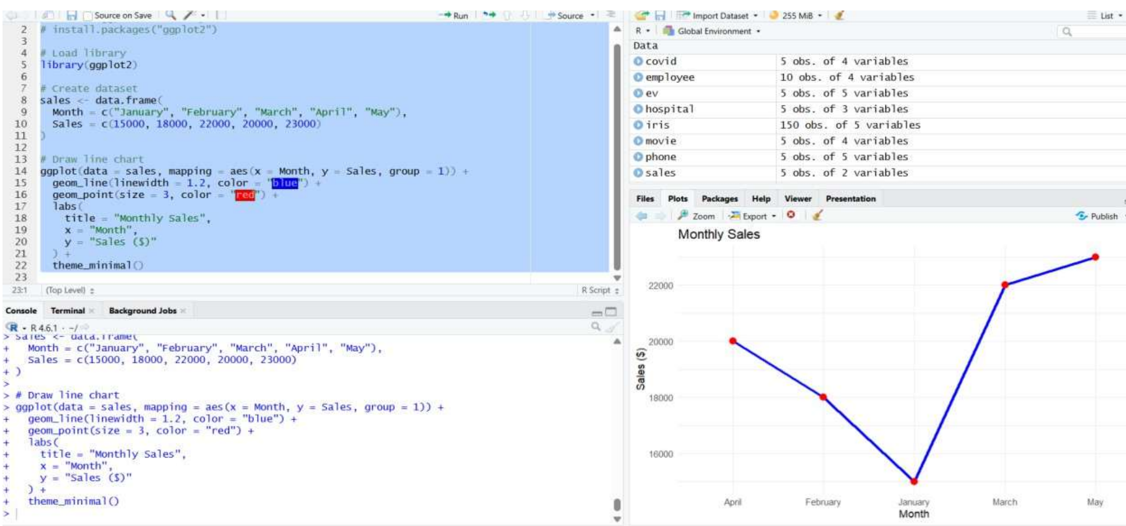
~Month, y

= ~Sales,

type =

"bar")

subplot(p1, p2, nrows = 2)




EXPERIMENT 2: Customer Feedback

Analysis

PROGRAM

**Step 1: Load Packages**

library(ggplot2)

library(wordcloud)

library(RColorBrewer)

**Step 2: Create Dataset**

```
customer <- data.frame(
  Age = c(20,25,30,35,40),
  Satisfaction = c("High","Medium","High","Low","Medium")
)

print(customer)
```

**Step 3: Histogram**

```
ggplot(customer, aes(x =
Age)) +
geom_histogram(binwidth =
5,
        fill = "skyblue",
        color = "black") +
 labs(
  title = "Customer Age
  Distribution", x = "Age",
  y = "Frequency"
 ) +
 theme_minima
l()
```

**Step 4: Pie Chart**

```
pie(
 table(customer$Satisfacti
 on), col = rainbow(3),
 main = "Customer Satisfaction Level"
)
```

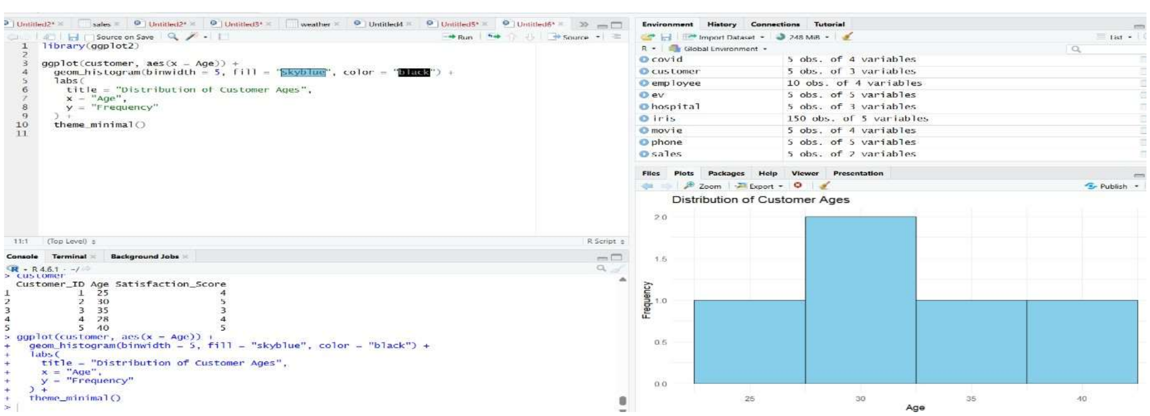
Step 5: Stacked Bar Chart

```
customer$Age_Group <- c("20-25","20-25","30-35","30-35","40+")
```

```
ggplot(customer,
    aes(x =
      Age_Group,  fill
      = Satisfaction)) +
  geom_bar() +
labs(
   title = "Satisfaction by Age
   Group", x = "Age Group",
   y = "Count"
  ) +
  theme_minimal()
```

**Step 6: Word Cloud** wordcloud(

```
  words =
  customer$Satisfaction, freq
  = c(2,2,1,1,2),
  colors = brewer.pal(8,"Dark2")
)
```



EXPERIMENT 3: Employee Performance Analysis

PROGRAM

**Step 1: Load Packages**

```
library(ggplot2)
library(plotly)
```

**Step 2: Create Dataset**

```
employee <- data.frame(
Employee_ID =
c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  Years_of_Service = c(5,3,7,4,2),
  Performance_Score = c(85,92,78,90,76)
)

print(employee)
```

Step 3: Line Chart

```
ggplot(employee,
      aes(x                        =
          Years_of_Service,  y
          =
          Performance_Score,
          group = 1)) +
  geom_line(color = "blue", linewidth =
  1.2) + geom_point(color = "red", size = 3)
  +
  labs(
    title = "Employee Performance
    Trend", x = "Years of Service",
```

```
  y = "Performance Score"
 ) +
 theme_minima
l()
```

**Step 4: Bar Chart**

```
ggplot(employe
e,
     aes(x = Department,
        fill          =
        Department)) +
 geom_bar() +
 labs(
  title = "Employees by
  Department", x =
  "Department",
  y = "Number of Employees"
 ) +
 theme_minimal()
```

**Step 5: Scatter Plot**

```
ggplot(employee,
     aes(x = Years_of_Service,
        y = Performance_Score)) +
 geom_point(size = 4, color = "blue") +
 geom_smooth(method = "lm",
          se = FALSE,
          color = "red") +
 labs(
  title = "Years of Service vs
  Performance", x = "Years of
```

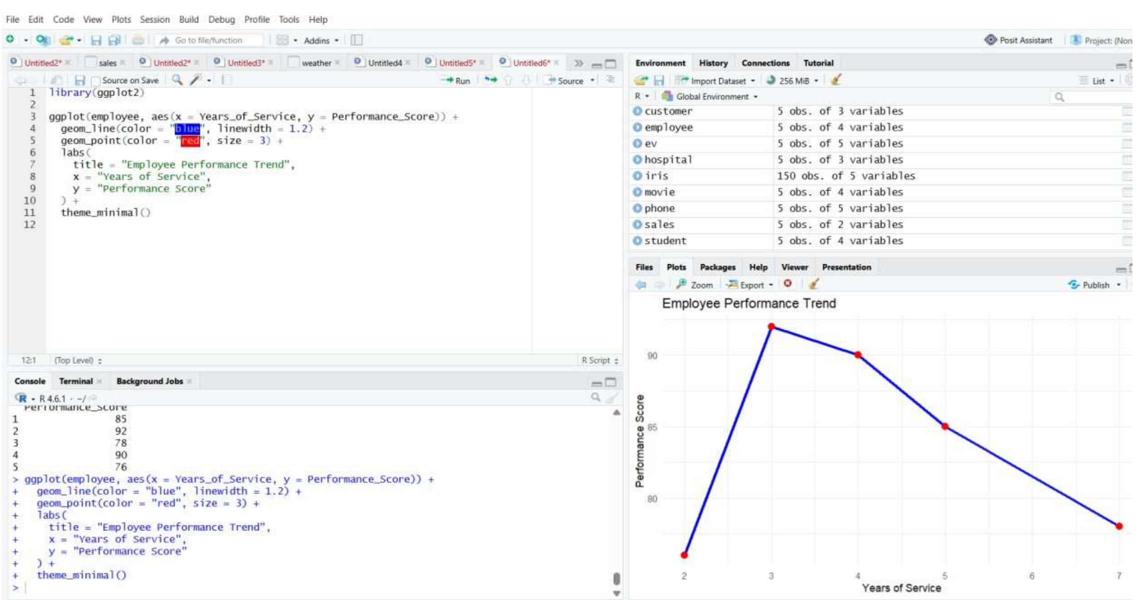
```
   Service",
   y = "Performance Score"
  ) +
  theme_minimal()
```

Step 6: Interactive Dashboard

```
p1 <- plot_ly(employee,
          x = ~Years_of_Service,
          y =
          ~Performance_Score,
          type = "scatter",
          mode = "lines+markers")

p2 <- plot_ly(employee,
          x = ~Department,
          type = "bar")

subplot(p1, p2, nrows = 2)
```



EXPERIMENT 4: Product Inventory

Analysis

PROGRAM

**Step 1: Load**

**Packages**

```
library(ggplot2)

library(plotly)
```

Step 2: Create Dataset

```
inventory <- data.frame(
  Product_Name = c("Product A","Product B","Product C","Product D","Product
  E"), Category =
  c("Electronics","Electronics","Furniture","Furniture","Furniture"),
  Price =
  c(500,650,450,700,550),
  Quantity =
  c(250,175,300,200,220)
)


print(inventory)
```

Step 3: Bar Chart

```
ggplot(inventory,
       aes(x =
           Product_Name, y
           = Quantity,
           fill                  =
  Product_Name))           +
  geom_bar(stat            =
  "identity") + labs(
    title      =      "Product
    Inventory",       x       =
```

"Product Name",
y = "Quantity"
) +
theme_minimal()

Step 4: Stacked Bar Chart

ggplot(inventory,
aes(x =
Category,
y = Quantity,
fill =
Product_Name)) +
geom_bar(stat =
"identity") + labs(
title = "Inventory by
Category", x = "Category",
y = "Quantity"
) +
theme_minimal()

Step 5: Scatter Plot

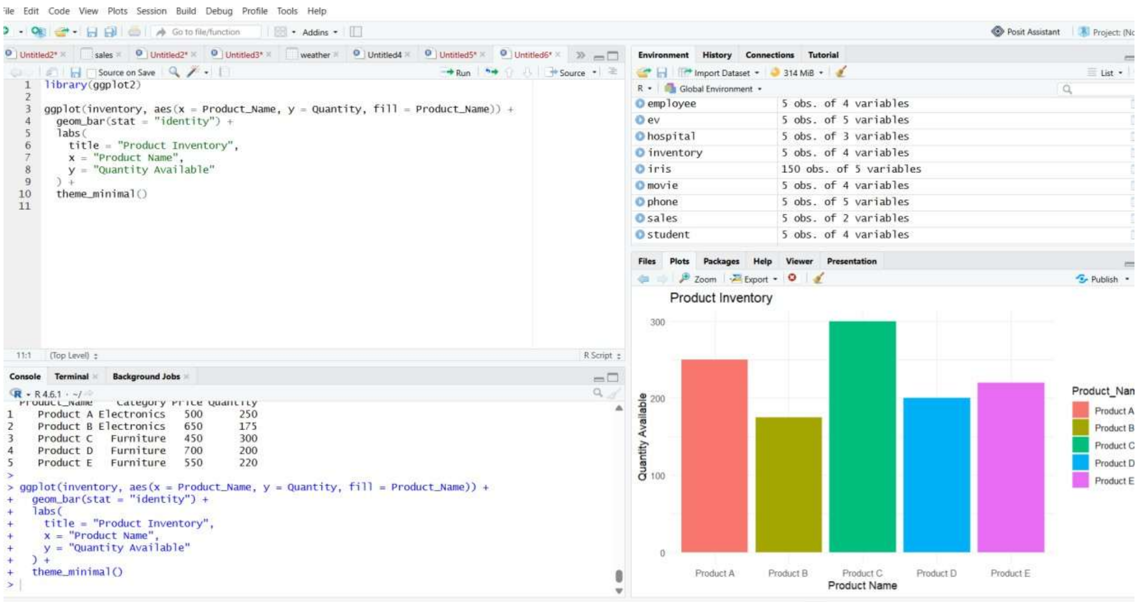
```
ggplot(inventory
     , aes(x =
     Price,
        y = Quantity)) +
  geom_point(size = 4, color =
  "blue") + geom_smooth(method
  = "lm",
           se = FALSE,
           color = "red") +
  labs(
   title = "Price vs
   Quantity", x = "Price",
   y = "Quantity"
  ) +
  theme_minimal()
```

Step 6: Interactive Dashboard

```
p1 <- plot_ly(inventory,
           x =
           ~Product_Name,
           y = ~Quantity,
           type = "bar")

p2 <- plot_ly(inventory,
           x = ~Price,
           y = ~Quantity,
           type = "scatter",
           mode =
           "markers")

subplot(p1, p2, nrows = 2)
```



SET 5: Website

PROGRAM

**Step 1: Load**

**Packages**

library(ggplot2)

library(plotly)

library(tidyr)

Step 2: Create Dataset

traffic <- data.frame(

Date =

as.Date(c(

"2023-01-01",

"2023-01-02",

"2023-01-03",

"2023-01-04",

"2023-01-05"

)),

Page_Views =
c( 1500,
1600,
1400,
1650,
1800
),

CTR = c(
2.3,
2.7,
2.0,
2.4,
2.6
)

)

print(traffic)

**Question 1**

**Line Chart**

ggplot(traffic,
aes(x = Date,
y =
Page_Views,
group = 1)) +

geom_line(color =

"blue", linewidth = 1.2)

+

geom_point(color =

"red", size = 3) +

labs(

title = "Daily Website Page

Views", x = "Date",

y = "Page Views"

) +

theme_minimal()